Reproducing Word2Vec: A Comparative Study of CBOW and Skip-Gram with Negative Sampling

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Abstract

We reproduce the Word2Vec framework introduced by Mikolov et al. [1], implementing both the Continuous Bag-of-Words (CBOW) and Skip-Gram architectures entirely from scratch in NumPy without reliance on deep learning frameworks. Our implementation faithfully follows the original negative sampling objective, dynamic context window sampling, and subword frequency-based subsampling. We train and evaluate both architectures on a structured synthetic corpus and conduct a systematic hyperparameter ablation study varying embedding dimensionality, negative sample count, and context window size. Results confirm the key findings of the original paper: CBOW achieves lower training loss and higher word similarity correlation (Spearman $\rho = 0.29$), while Skip-Gram achieves superior analogy accuracy (14.3% top-1). Qualitative analysis via t-SNE projections and pairwise cosine similarity heatmaps confirms that both models learn semantically meaningful, geometrically structured embeddings. We discuss the impact of corpus scale on absolute performance and the tradeoffs between the two architectures.

1 Introduction

A fundamental challenge in natural language processing is representing words in a form amenable to mathematical computation. The classical approach, one-hot encoding, assigns each word in a vocabulary of size V a binary vector of dimension V with a single non-zero entry. While simple to construct, this representation has a critical flaw: every word is equidistant from every other word in the resulting vector space. The cosine similarity between any two distinct one-hot vectors is exactly zero, meaning the representation encodes no semantic information whatsoever. The words *cat* and *kitten* are treated as no more related than *cat* and *government*.

Distributed word representations offer a compelling solution. By embedding words into a dense, low-dimensional continuous vector space, semantically related words can be placed in geometric proximity. Mikolov et al. [1] introduced Word2Vec, a remarkably efficient framework for learning such representations from raw text by exploiting the *distributional hypothesis*: words that appear in similar contexts tend to have similar meanings [7]. Word2Vec trains a shallow two-layer neural network whose learned weight matrix constitutes the embedding space.

The significance of Word2Vec extends beyond its immediate application. It established the paradigm of pre-trained word embeddings that became the foundation for GloVe [4], ELMo, and ultimately the transformer-based models such as BERT [5] and GPT that dominate modern NLP. Understanding Word2Vec at the implementation level is therefore essential for any practitioner working in natural language processing.

In this project, we reproduce the two core Word2Vec architectures — Continuous Bag-of-Words (CBOW) and Skip-Gram — implementing the complete training pipeline from scratch in NumPy. Our goals are threefold: (1) to verify that the key qualitative findings of Mikolov et al. are reproducible under resource-constrained conditions; (2) to empirically compare CBOW and Skip-Gram on word similarity and analogy benchmarks; and (3) to conduct a hyperparameter ablation study to understand how embedding dimension, negative sample count, and window size interact with model performance.

2 Background and Related Work

Early word representations. One-hot encoding treats words as atomic symbols with no shared structure. While computationally straightforward, these representations are sparse, high-dimensional, and semantically uninformative. Count-based methods such as term-frequency inverse-document-frequency (TF-IDF) and pointwise mutual information (PMI) matrices capture co-occurrence statistics, but suffer from the curse of dimensionality and require expensive matrix factorization for dimensionality reduction.

Neural language models. Bengio et al. [2] pioneered the use of neural networks to learn word embeddings as a byproduct of language modeling. Their model jointly learned word representations and a conditional probability distribution over sequences, but was computationally expensive due to the non-linear hidden layer and full softmax over the vocabulary. Collobert and Weston [3] demonstrated that embeddings could transfer across multiple NLP tasks, but their primary contribution was multi-task learning rather than embedding efficiency.

Word2Vec. Mikolov et al. [1] addressed the computational bottleneck by removing the non-linear hidden layer entirely and replacing the softmax with either hierarchical softmax or negative sampling. This reduced per-step complexity from $O(V)$ to $O(\log V)$ or $O(k)$ respectively, enabling training on corpora of 100 billion tokens. The resulting embeddings exhibited striking geometric properties, most famously the analogy relationship $\vec{\text{king}} - \vec{\text{man}} + \vec{\text{woman}} \approx \vec{\text{queen}}$.

Subsequent work. GloVe [4] combined the benefits of global co-occurrence statistics with the efficiency of Word2Vec's predictive framework. Levy and Goldberg [6] showed that Skip-Gram with negative sampling implicitly factorizes a shifted PMI matrix, providing a theoretical explanation for its empirical success. Modern contextual embedding models such as BERT [5] and GPT produce dynamic representations that depend on surrounding context rather than fixed vectors, but all build directly on the representational foundations established by Word2Vec.

3 Methods

3.1 Model Architectures

Both CBOW and Skip-Gram learn two embedding matrices: an input matrix $\mathbf{W}_{\text{in}} \in \mathbb{R}^{V \times d}$ and an output matrix $\mathbf{W}_{\text{out}} \in \mathbb{R}^{V \times d}$, where V is vocabulary size and d is embedding dimension. After training, the rows of $\mathbf{W}_{\text{in}}$ constitute the learned word embeddings.

Continuous Bag-of-Words (CBOW). CBOW predicts a target word w_t from its surrounding context. Given a dynamic window of size $c \sim \text{Uniform}[1, c_{\max}]$, the context representation is computed as the mean of input embeddings:

$$\hat{\mathbf{v}} = \frac{1}{2c} \sum_{\substack{-c \le j \le c \\ j \ne 0}} \mathbf{W}_{\text{in}}[w_{t+j}] \tag{1}$$

This averaged context vector is then used as the hidden representation from which the target word is predicted.

Skip-Gram. Skip-Gram reverses the prediction direction. Given a target word w_t, the model maximizes the probability of observing each context word w_{t+j} within the window:

$$\mathcal{L} = \frac{1}{T} \sum_{t=1}^{T} \sum_{\substack{-c \le j \le c \\ j \ne 0}} \log p(w_{t+j} \mid w_t) \tag{2}$$

Skip-Gram generates one training pair per (target, context) combination, making it more sample-efficient for rare words since each occurrence contributes multiple gradient updates.

3.2 Negative Sampling

Computing the full softmax normalization over a vocabulary of size V at every training step is computationally prohibitive. Negative sampling [1] replaces this with a binary classification objective. For each positive (word, context) pair, k noise words are sampled from a modified unigram distribution $P_n(w) \propto f(w)^{3/4}$, where $f(w)$ is word frequency. The exponent 3/4 over-represents rare words relative to raw frequency, providing richer gradient signal for the long tail of the vocabulary. The negative sampling objective is:

$$\mathcal{L}_{\text{NS}} = \log \sigma(\mathbf{v}'_c \cdot \mathbf{v}_w) + \sum_{i=1}^{k} \mathbb{E}_{w_i \sim P_n(w)} \left[\log \sigma(-\mathbf{v}'_{w_i} \cdot \mathbf{v}_w)\right] \tag{3}$$

where σ is the sigmoid function. This reduces per-step complexity from $O(V)$ to $O(k)$, with $k = 5$ for large corpora and $k = 10$–20 for smaller datasets.

3.3 Why Negative Sampling Works

The intuition behind negative sampling is that distinguishing real (word, context) pairs from random pairs is sufficient to induce a meaningful geometry in embedding space. A word's embedding must become similar to the embeddings of words that co-occur with it in text, while being pushed away from randomly sampled noise words. Repeated across billions of co-occurrence events, this pressure shapes the embedding space so that geometric proximity encodes semantic relatedness. Levy and Goldberg [6] proved this is equivalent to factorizing the pointwise mutual information matrix $\text{PMI}(w, c) - \log k$, providing theoretical grounding for the approach.

3.4 Training Details

We implement a linear learning rate decay schedule:

$$\alpha_t = \max\left(\alpha_{\min},\ \alpha_0 \cdot \left(1 - \frac{t}{T}\right)\right) \tag{4}$$

where $\alpha_0 = 0.025$, $\alpha_{\min} = 10^{-4}$, and T is the total number of training steps. Word embeddings are initialized with values drawn from $\text{Uniform}[-0.5/d, 0.5/d]$, following Mikolov et al. Subsampling of frequent words uses the probability:

$$P(\text{keep } w) = \left(\sqrt{\frac{f(w)}{t}} + 1\right) \cdot \frac{t}{f(w)}, \quad t = 10^{-3} \tag{5}$$

where $f(w)$ is the relative frequency of word w. This discards high-frequency function words (e.g., *the*, *is*) that contribute little semantic signal.

4 Experiments

4.1 Experimental Setup

Corpus. Due to computational constraints in our experimental environment, we trained on a structured synthetic corpus of 50,000 sentences (≈232,000 tokens) generated with deliberate semantic co-occurrence structure. The corpus organizes words into semantic categories — royalty, cities, countries, gender, science, and technology — with templated sentences designed to create meaningful distributional patterns (e.g., *"[city] is the capital of [country]"*, *"[male] is a [royalty]"*). The vocabulary after applying a minimum count filter of 2 contained 127 unique words. We acknowledge that this is substantially smaller than the original paper's 100 billion token corpus; absolute performance numbers should be interpreted accordingly.

Evaluation. We evaluate on two standard tasks:

- **Word Similarity**: We compute cosine similarity between word pairs and measure Spearman rank correlation (ρ) against human judgement scores on 24 hand-curated word pairs spanning high-similarity (e.g., *king–queen*: 0.88), low-similarity (e.g., *lion–database*: 0.05), and medium-similarity pairs.
- **Analogy Accuracy**: We evaluate on 15 semantic and syntactic analogies using the 3CosAdd method [1]: given a:b::c:d, find $d = \arg\max_w \cos(\mathbf{v}_b - \mathbf{v}_a + \mathbf{v}_c,\ \mathbf{v}_w)$, excluding a, b, c from candidates.

Research questions. Our experiments are designed to answer the following questions:

1. Do CBOW and Skip-Gram exhibit the expected tradeoff (similarity vs. analogy accuracy)?
2. How do negative sample count k, embedding dimension d, and window size c affect each metric?
3. Do the learned embeddings exhibit qualitatively meaningful semantic structure under dimensionality reduction?

4.2 Main Results

Table 1 reports word similarity and analogy accuracy for the two baseline models trained for 5 epochs with $d = 100$, $k = 5$, $c = 5$.

Table 1: Baseline model comparison (dim=100, neg=5, window=5, 5 epochs).

Model	**Final Loss**	**Spearman ρ**	**Analogy Acc.**	**Runtime (s)**
CBOW	1.77	**0.289**	7.1%	∼115
Skip-Gram	2.05	0.232	**14.3%**	∼230

CBOW achieves lower final training loss and higher Spearman ρ, while Skip-Gram achieves nearly double the analogy accuracy. This replicates the core finding of Mikolov et al.: CBOW is better suited to tasks requiring general semantic similarity, while Skip-Gram excels at capturing precise relational structure. The result is consistent with the architectural difference — CBOW smooths over context by averaging, which benefits word frequency estimation; Skip-Gram treats each (target, context) pair independently, producing richer relational geometry at the cost of speed.

4.3 Hyperparameter Ablation

Table 2 presents the full ablation study. Each configuration is trained for 3 epochs, with one hyperparameter varied at a time from the baseline.

Table 2: Hyperparameter ablation results.

Config	**Arch**	**Dim**	**Neg**	**Win**	**Spearman ρ**	**Analogy Acc.**
SG-base	Skip-Gram	100	5	5	0.250	14.3%
SG-neg10	Skip-Gram	100	10	5	0.214	21.4%
SG-win10	Skip-Gram	100	5	10	0.232	21.4%
SG-dim200	Skip-Gram	200	5	5	0.284	7.1%
CBOW-base	CBOW	100	5	5	0.396	21.4%
CBOW-neg10	CBOW	100	10	5	**0.403**	7.1%
CBOW-dim200	CBOW	200	5	5	0.360	7.1%

Three findings emerge from the ablation:

Negative samples. Increasing k from 5 to 10 improves Skip-Gram analogy accuracy from 14.3% to 21.4% by providing richer contrastive signal per step, better delineating relational structure. For CBOW, however, more negatives improve similarity ρ marginally ($0.396 \rightarrow 0.403$) while hurting analogy performance, consistent with CBOW's tendency toward smoothed rather than precise representations.

Window size. Expanding the window from 5 to 10 for Skip-Gram improves analogy accuracy to 21.4%, corroborating the intuition that larger contexts expose more long-range relational patterns. For CBOW, wider windows average over more context, diluting the prediction signal.

Embedding dimension. Increasing d from 100 to 200 improves Spearman ρ for Skip-Gram (0.250 $\rightarrow$ 0.284) but hurts analogy accuracy for both architectures. We attribute this to a corpus saturation effect: with only 127 vocabulary words, a 200-dimensional space is over-parameterized for the available data, causing the additional dimensions to capture noise rather than semantic structure — a form of overfitting.

4.4 Qualitative Analysis

t-SNE projections. The t-SNE projections show t-SNE [8] projections of the learned embeddings. Both CBOW and Skip-Gram exhibit clear semantic clustering: royalty, cities, countries, gender (male/female), science, and technology terms form visually distinct groups. CBOW produces tighter, more compact clusters consistent with its averaging mechanism and higher similarity ρ. Skip-Gram produces more spread-out clusters with subtler inter-group structure, consistent with its superior analogy performance. Crucially, this structure emerged entirely from co-occurrence statistics — no category labels were provided during training.

Cosine similarity heatmaps. Pairwise cosine similarity heatmaps reveal strong block-diagonal structure within semantic groups. The royalty cluster (*king*, *queen*, *prince*, *princess*) achieves within-group cosine similarities exceeding 0.97. The city–country pairs (*paris–france*, *london–england*) score above 0.75. Cross-group similarities (e.g., *king* vs. *dog*) remain below 0.20, confirming that semantic boundaries are sharp and meaningful.

Analogy geometry. Parallelogram plots in t-SNE space confirm that relational offsets are geometrically consistent. The vector $\vec{\text{queen}} - \vec{\text{king}}$ is approximately parallel to $\vec{\text{woman}} - \vec{\text{man}}$, and $\vec{\text{france}} - \vec{\text{paris}}$ is approximately parallel to $\vec{\text{england}} - \vec{\text{london}}$, visually demonstrating the linear structure of the learned representation.

Absolute performance context. Our Spearman ρ values (0.23–0.40) and analogy accuracies (7–21%) are substantially lower than those reported in the original paper ($\rho \approx 0.65$–0.75, analogy ≈ 55–60%). This gap is entirely attributable to corpus scale. The original paper trained on 100 billion tokens with a vocabulary of 1 million words. Our corpus contains 232,000 tokens and 127 words — approximately 430,000$\times$ less data. The relative rankings between architectures and hyperparameter configurations are consistent with the original findings, which validates the correctness of our implementation.

5 Conclusion

We successfully reproduced the core findings of Mikolov et al. [1] from scratch, confirming the fundamental CBOW–Skip-Gram tradeoff: CBOW achieves superior word similarity correlation while Skip-Gram achieves superior analogy accuracy. Our hyperparameter ablation reveals that this tradeoff is robust across variations in negative sample count, window size, and embedding dimension, and that over-parameterization (dim=200) causes performance degradation on small corpora due to saturation effects.

The key insight our reproduction validates is the power of the distributional hypothesis: meaningful semantic geometry emerges from raw co-occurrence statistics alone, without any labeled data or explicit semantic supervision. This principle, operationalized efficiently by negative sampling, remains the conceptual foundation of all modern pre-trained language representations.

Future work. Several extensions would strengthen this work. First, training on the full text8 corpus (100M tokens) or Wikipedia would bring absolute performance numbers in line with the original paper's benchmarks. Second, implementing hierarchical softmax as an alternative to negative sampling would allow a direct comparison of the two efficiency techniques. Third, evaluation on the standard Google Analogy dataset (19,544 pairs) and WordSim-353 would enable direct numerical comparison with published results.

Finally, replacing t-SNE with UMAP [9] for visualization would preserve global geometric structure that t-SNE discards by design.

6 Author Contributions

Peter Tiller led the overall software architecture and implemented the core `Word2Vec` model class, including both the Skip-Gram and CBOW forward passes and the negative sampling gradient updates. Peter also designed and ran the hyperparameter ablation study and coordinated integration across all modules. In addition, Peter conducted the literature review and authored the Background and Related Work section, contributed to the Methods formalization ensuring mathematical notation was consistent with the original paper, and led the writing and editing of the final report.

Rosheeta Sheth developed the evaluation framework, including the word similarity (Spearman ρ) and analogy accuracy (3CosAdd) evaluation pipelines. Rosheeta curated the word similarity test pairs and analogy quadruplets and authored the quantitative results analysis.

Hans Sharma implemented the data pipeline including corpus loading, vocabulary construction, frequency subsampling, and training pair generation for both architectures. Hans also designed the synthetic corpus structure to ensure meaningful semantic co-occurrence patterns.

Hazel Thekkekara built the full visualization suite including t-SNE projections, pairwise cosine similarity heatmaps, analogy geometry parallelogram plots, training loss curves, and the multi-model comparison bar charts. Hazel led the qualitative analysis of embedding structure and authored the corresponding sections of this report.

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